

Department of Computer Science, BRACU
Computer Networks, CSE 421
QUIZ 4, SET A
Time 20 min, Marks: 15

ID	Name:	Sec
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1.	Left_Network PC1 PC2 PC3 PC4 PC5 Printer0 Switch0 Right_Network PC6 PC7 PC8 PC9 PC10 Server0 Switch1 Router0 F0/0 F0/1 192.168.1.0/24 192.168.2.0/24 <table> <thead> <tr> <th>Device</th><th>IP</th><th>Device</th><th>IP</th></tr> </thead> <tbody> <tr> <td>PC1</td><td>192.168.1.3</td><td>PC6</td><td>192.168.2.101</td></tr> <tr> <td>PC2</td><td>DHCP</td><td>PC7</td><td>192.168.2.100</td></tr> <tr> <td>PC3</td><td>DHCP</td><td>PC8</td><td>DHCP</td></tr> <tr> <td>PC4</td><td>DHCP</td><td>PC9</td><td>DHCP</td></tr> <tr> <td>PC5</td><td>DHCP</td><td>PC10</td><td>DHCP</td></tr> <tr> <td>Printer0</td><td>192.168.1.2</td><td>Server0</td><td>192.168.2.102</td></tr> <tr> <td>F0/0</td><td>192.168.1.1</td><td>F0/1</td><td>192.168.2.1</td></tr> </tbody> </table> <pre> int f0/0 ip address dhcp -> fetch interface IP via DHCP ip exclude 192.168.1.1 192.168.1.2 pool left_network network 192.168.1.0 255.255.0.0 default-gateway 192.168.1.1 </pre> a. Identify any 2 issues in the above configuration and fix it. ANS: Wrong subnet mask in DHCP pool. The Left Network is /24, not /16. Pc 1 is not excluded b. After solving the issues in (a), clients in Left-Network are complaining that they are still not receiving IPs. What's the solution? Be fully specific. Assume Server0 = DHCP. ANS: Router0(config)# interface f0/0 Router0(config-if)# ip helper-address 192.168.2.102	Device	IP	Device	IP	PC1	192.168.1.3	PC6	192.168.2.101	PC2	DHCP	PC7	192.168.2.100	PC3	DHCP	PC8	DHCP	PC4	DHCP	PC9	DHCP	PC5	DHCP	PC10	DHCP	Printer0	192.168.1.2	Server0	192.168.2.102	F0/0	192.168.1.1	F0/1	192.168.2.1	2+3 =5
Device	IP	Device	IP																															
PC1	192.168.1.3	PC6	192.168.2.101																															
PC2	DHCP	PC7	192.168.2.100																															
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Printer0	192.168.1.2	Server0	192.168.2.102																															
F0/0	192.168.1.1	F0/1	192.168.2.1																															
2.	Given the following output from a "show ip route" command. <pre> S 10.0.0.0/8 [1/0] via 172.16.40.2 C 64.100.0.0/16 is directly connected, Serial0/1 C 128.107.0.0/16 is directly connected, Serial0/2 S 172.16.40.0/24 [5/0], Serial0/0 S 172.16.0.0/16 [5/0], Serial0/0 S 172.16.0.0/24 [50/0], Serial1/0 C 192.168.1.0/24 is directly connected, FE0/0 S 192.168.2.0/24 [1/0] via 172.16.40.2 C 198.133.219.0/24 is directly connected, FE0/1 </pre> a. A packet is destined for the network 172.16.4.0/24. What action will the router take? If the packet is dropped, how can you handle it? Drop, no entry of default route b. What's the exit interface port of the route set to 192.168.2.0/24? Serial0/0 c. Write the command the network administrator had to issue when configuring the 172.16.0.0/24 network. ip route 172.16.0.0 255.255.255.0 Serial1/0	4+2 +4 =10																																

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1.

Left_Network

Right_Network

Router0

Switch0

Switch1

PC1, PC2, PC3, PC4, PC5, Printer0

PC6, PC7, PC8, PC9, PC10, Server0

192.168.1.0/24

192.168.2.0/24

Device	IP	Device	IP
PC1	192.168.1.3	PC6	192.168.2.101
PC2	DHCP	PC7	192.168.2.100
PC3	DHCP	PC8	DHCP
PC4	DHCP	PC9	DHCP
PC5	DHCP	PC10	DHCP
Printer0	192.168.1.2	Server0	192.168.2.102
F0/0	192.168.1.1	F0/1	192.168.2.1

```

int f0/0
ip address dhcp -> fetch interface IP via DHCP

ip exclude 192.168.1.1 192.168.1.2
pool left_network
network 192.168.1.0 255.255.0.0
default-gateway 192.168.1.1

```

c. Identify any 2 issues in the above configuration and fix it. **ANS:**

d. After solving the issues in (a), clients in Left-Network are complaining that they are still not receiving IPs. What's the solution? Be fully specific. Assume Server0 = DHCP. **ANS:**

2+3
=5

2.

Given the following output from a "show ip route" command.

```

S    10.0.0.0/8 [1/0] via 172.16.40.2
C    64.100.0.0/16 is directly connected, Serial0/1
C    128.107.0.0/16 is directly connected, Serial0/2
C    172.16.40.0/24 [10/0], Serial0/0
C    192.168.1.0/24 is directly connected, FE0/0
S    192.168.2.0/24 [5/0] via 172.16.40.2
C    198.133.219.0/24 is directly connected, FE0/1
S*   0.0.0.0 0.0.0.0 via 172.16.40.2

```

a. Identify if you can for sure say there's a back up/ floating route configured Above.
We cannot say for sure that a floating/backup route is configured, except that a default static route exists.

b. Can the above route handle packets which are destined for network addresses unknown to the above router?
Yes, unknown destination packets are forwarded using the default route.

c. Write the command the network administrator had to issue when configuring the 192.168.2.0/24 network.
ip route 192.168.2.0 255.255.255.0 172.16.40.2 5

4+2
+4=
10

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1. a.	PC1 broadcasts a DHCP Discover packet. Identify if it can reach R3 <u>with one line</u> explanation. Ans:	2
b.	How can you configure the above topology such that the packet can reach the R3? Write the commands listed below: Router name? _____ R2 _____ Router interface? _____ g1 _____ IP Helper address? _____ 192.168.2.1 _____	3
2.	Given the following output from a "show ip route" command. <pre>S 10.0.0.0/8 [1/0] via 172.16.40.2 C 64.100.0.0/16 is directly connected, Serial0/1 C 128.107.0.0/16 is directly connected, Serial0/2 S 172.16.40.0/24 [5/0], Serial0/0 S 172.16.0.0/16 [5/0], Serial0/0 S 172.16.0.0/24 [50/0], Serial1/0 C 192.168.1.0/24 is directly connected, FE0/0 S 192.168.2.0/24 [1/0] via 172.16.40.2 C 198.133.219.0/24 is directly connected, FE0/1</pre> d. A packet is destined for the network 172.16.4.0/24. What action will the router take? If the packet is dropped, how can you handle it? e. What's the exit interface port of the route set to 192.168.2.0/24? f. Write the command the network administrator had to issue when configuring the 172.16.0.0/24 network.	4+2 +4 =10

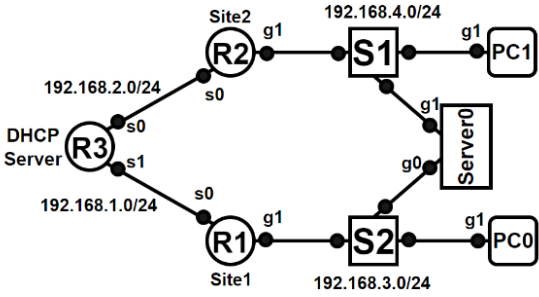
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1. a.	 PC0 broadcasts a DHCP Discover packet. Identify if it can reach R3 with <u>one line explanation</u>. Ans: No—PC1's DHCP Discover is a Layer-2 broadcast and routers (R2/R1) do not forward broadcasts unless a DHCP relay (ip helper-address) is configured.	2+3 =5
b.	How can you configure the above topology such that the packet can reach the R3? Write the commands listed below: Router name? _____ R1 _____ Router interface? _____ g1 _____ IP Helper address? _____ 192.168.1.1 _____	

2.	Given the following output from a "show ip route" command. <pre> S 10.0.0.0/8 [1/0] via 172.16.40.2 C 64.100.0.0/16 is directly connected, Serial0/1 C 128.107.0.0/16 is directly connected, Serial0/2 C 172.16.40.0/24 [10/0], Serial0/0 C 192.168.1.0/24 is directly connected, FE0/0 S 192.168.2.0/24 [5/0] via 172.16.40.2 C 198.133.219.0/24 is directly connected, FE0/1 S* 0.0.0.0 0.0.0.0 via 172.16.40.2 </pre> d. Identify if you can for sure say there's a back up/ floating route configured Above. We cannot say for sure that a floating/backup route is configured, except that a default static route exists. e. Can the above route handle packets which are destined for network addresses unknown to the above router? Yes, unknown destination packets are forwarded using the default route. f. Write the command the network administrator had to issue when configuring the 192.168.2.0/24 network. ip route 192.168.2.0 255.255.255.0 172.16.40.2 5	4+2 +4= 10
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